Survival Analysis of Faculty Retention in STEM

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Background Information

- Faculty retention is a crucial issue in academia, particularly in STEM disciplines.

- Understanding how long faculty members remain in tenure-track positions and identifying factors that influence retention can provide valuable insights for universities aiming to improve faculty retention policies.

- Questions we aim to answer:

- Do Ivy-League schools have higher retention rates than non Ivy League schools?
- What departments generally have higher or lower retention rates?
- Does gender have any role in retention rates?

Data and Methodology

- Observational Study: Data were collected from publicly available sources (college catalogs, departmental websites) to track faculty careers over time.
- Experimental/Observational Units: Individual assistant professors hired in STEM fields since 1990.

Sample Size

- Total Sample: 525 faculty members from 14 different schools
- Subgroups:
  - Gender: Men vs. Women.
  - Discipline: Engineering, Mathematics, Biology, Chemistry, etc. We later aggregated these by letter in ‘STEM’.
  - Ivy League vs. Non-Ivy League: Columbia, Cornell, Princeton vs. other universities in the dataset.

Data Collection

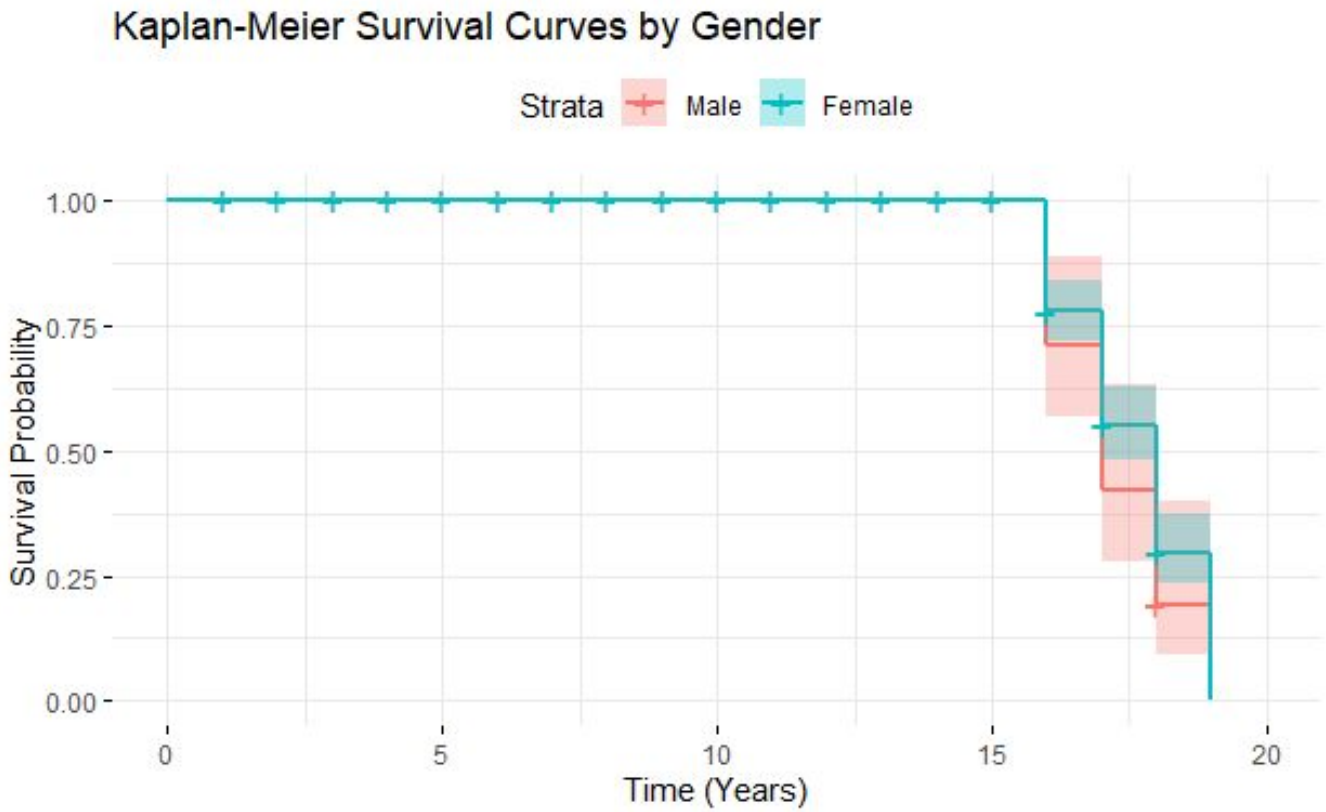
- Sources: College catalogs, departmental websites, and other public records.
- Challenges/Bias:
  - Data were limited to publicly available information, which may not represent all departures or reasons for leaving.
  - Potential bias in self-reported or incomplete records.

Variables Collected

- Response Variable: Time to departure (years in tenure track).
- Covariates/Factors: Gender, discipline, university type (Ivy League vs. Non-Ivy League), STEM category.

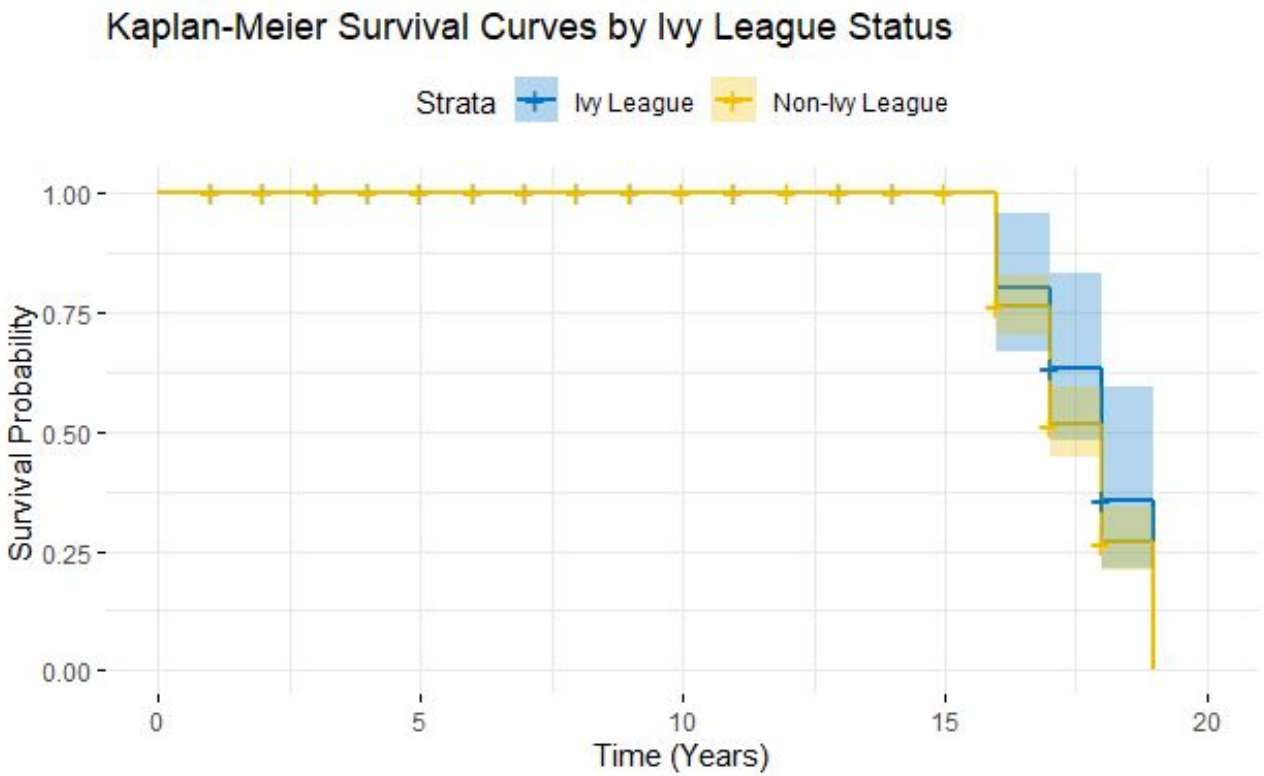
Results and Analysis

- Log-Rank Test: p = 0.2, which implies that there is no significant difference between Male and Female retention
- Only 91 female faculty members vs 434 male faculty members, so it isn't a super even difference between the two



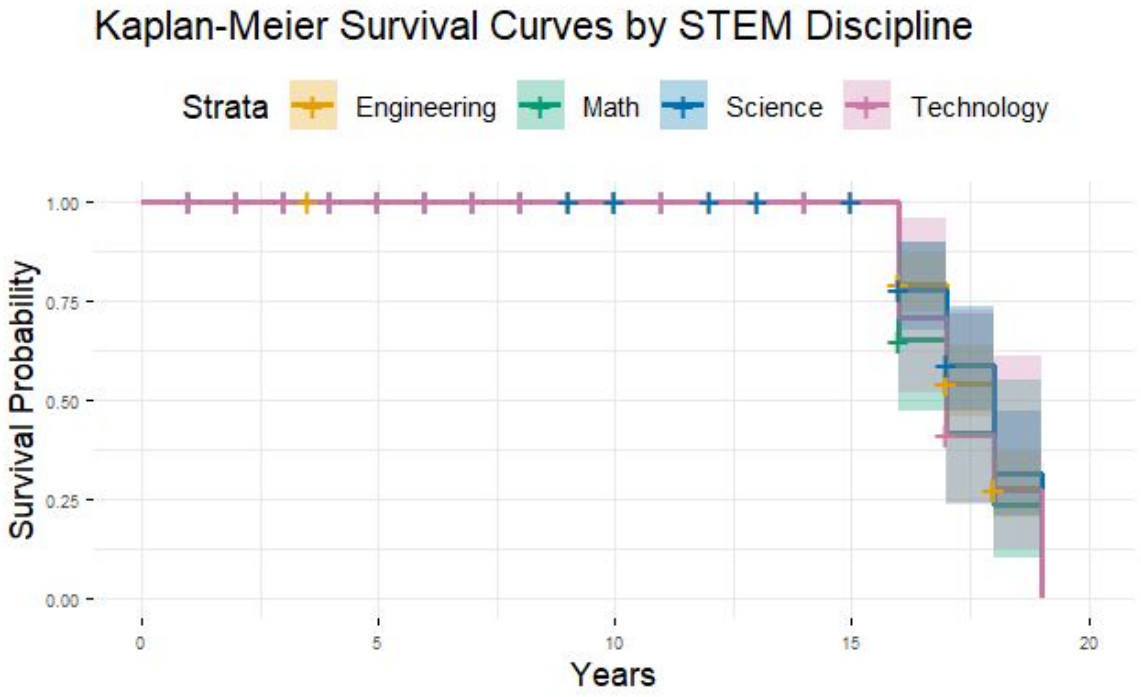
Results and Analysis

- Faculty at Ivy League universities show slightly longer median survival times than those at Non-Ivy institutions.
- Log-Rank Test: p = 0.3, which implies that there is no significant difference between Ivy League and Non-Ivy League retention.



Results and Analysis

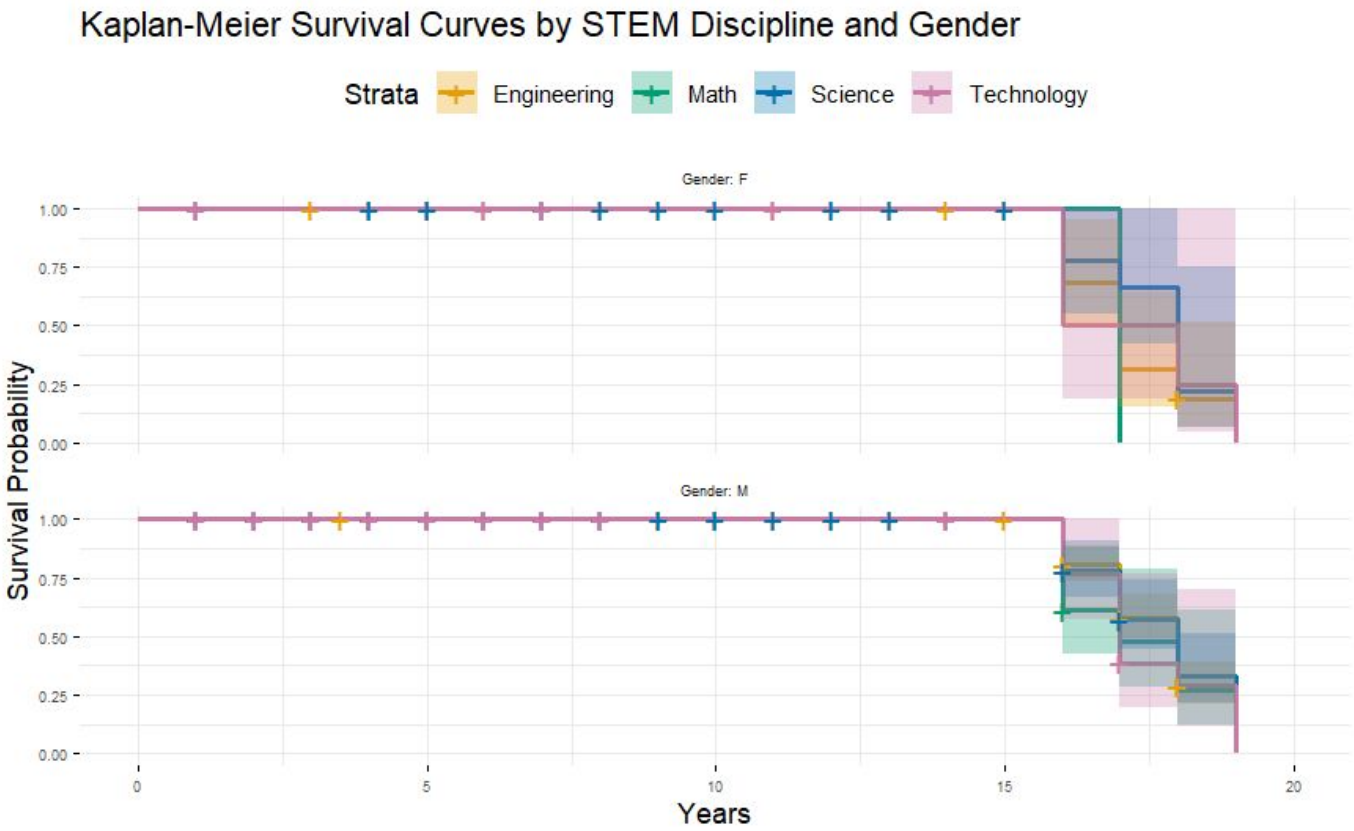
- Faculty show minimal differences in retention between STEM categories
- Log-Rank Test: p = 0.7, which implies that there is no significant difference in retention between STEM categories
- Large sample sizes for each group, even though there were 14 types of engineering, while Technology and Math had one field each.



Results and Analysis

In a survival curve that accounts for STEM subcategory and Gender, female Math and CS professors appear to leave slightly earlier than their male peers.

There were only 6 and 12 professors in each respective category, so this could easily be a sample size issue.



Results and Analysis

Here's a bit of descriptive statistics for you

For Departments

- Minimum Survival Time was 1 year (AGR, BIOL, CSCI, MATH, PHYS, etc.)
- Mean Survival Time was about 11.38 years
- Maximum Survival Time was 19 years
- Ocean Engineering had the highest* retention rate with Astronomy being the runner up, and Nuclear Engineering had the lowest

For Gender:

- Minimum Survival Time was 1 year
- Mean Survival Time was roughly 10.988 years for male faculty members 10.967 for female faculty members
- Maximum Survival Time was 19 years

For Ivy-League Status:

- Minimum Survival Time was 1 year
- Mean Survival Time was roughly 8.867 years for Ivy League schools, 11.471 years for Non-Ivy-League schools
- Maximum Survival Time was 19 years

For STEM Categories:

- Minimal survival time was 1 year
- Mean survival time was 9 for math, 10 for technology, 11 for engineering, and 11.5 for science.
- Maximum survival time was 19 years.
- When accounting for gender, female professors had a lower survival time in all subcategories except Science.

Conclusions

Main Takeaways

- Overall Retention: Mean tenure-track duration varies significantly by department.
- Gender Parity: Men and women experience similar retention rates, with no significant differences in survival curves.
- Retention varies considerably across disciplines, likely due to field-specific challenges and institutional policies.
- However, variation is obscured when aggregated into STEM fields, likely because there is a lot of variation within these fields

Limitations

- Data Source: Reliance on publicly available records may miss some departures or reasons for leaving.
- Censoring: Some faculty were still in their positions at the time of data collection (2009), leading to right-censored data.

Future Steps

- Further Analysis:
  - Investigate reasons for higher attrition in each department.
  - Explore the impact of institutional policies on retention.
  - Analyze retention by additional covariates (e.g., funding, mentorship).
- Policy Implications:
  - Develop targeted retention strategies for high-attrition disciplines.
  - Address potential disparities in faculty retention.

Any Questions?